

Tap Tone Lab Report (Template Demo)

Operational Acoustic-Structural Mapping (Speaker Drive, Roving Measurement)

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Date	2025-12-26 00:30:01
Experiment ID	EXP-2025-12-25-0001
Instrument ID	OM-Prototype-01
Build Stage	Top braced, no finish
Bundle Path	/mnt/data/tap_tone_lab_report_template/sample_bundle
Units	mm

Abstract

This report summarizes an operational roving measurement of an acoustic guitar driven by a reproducible speaker-air excitation. A fixed reference microphone and a roving microphone were used to estimate complex transfer behavior and coherence over a defined spatial grid. Results include resonance features, coherence-based confidence, and spatial localization indicators to support wolf-region identification.

1. Introduction

Goal: achieve high observability of coupled vibro-acoustic behavior with minimal intrusion (no added plate mass). This report documents apparatus, protocol, signal processing, results, and an error/confidence framework suitable for repeatable shop or lab use.

2. Physical System Description

Instrument ID	OM-Prototype-01
Build Stage	Top braced, no finish
Strings	n/a
Support Condition	n/a
Temperature	n/a
Humidity	n/a

2.2 Measurement Grid

Grid points: 4. Units: mm.

Label	x (mm)	y (mm)
P01	0.0	0.0
P02	50.0	0.0
P03	0.0	60.0
P04	-40.0	60.0

3. Experimental Apparatus

Excitation	speaker-air drive
Signal	log chirp
Drive Range (Hz)	[30, 2000]
Reference Mic	Generic measurement mic
Roving Mic	Generic measurement mic
Sample Rate (Hz)	48000
Duration (s)	6.0

4. Measurement Protocol

Roving mic moved point-to-point across a defined grid; reference mic fixed; 3 repeats per point.

5. Signal Processing

Operational transfer estimate uses cross/auto spectral densities: $H_{ir}(f)=G_{ir}(f)/G_{rr}(f)$. Coherence is used as a validity gate. For time-gated impulse response workflows, a deconvolution step yields $h(t)$, followed by windowing $h_g(t)$ prior to FFT.

6. Results

6.1 Wolf-Region Indicators

Point	x (mm)	y (mm)	Localization	Coherence
P03	0.0	60.0	2.850	0.960
P01	0.0	0.0	2.100	0.930
P02	50.0	0.0	1.400	0.880
P04	-40.0	60.0	1.100	0.720

Interpretation: high localization combined with high coherence suggests spatially concentrated resonant behavior and is a candidate indicator of wolf-prone regions when correlated with playing tests.

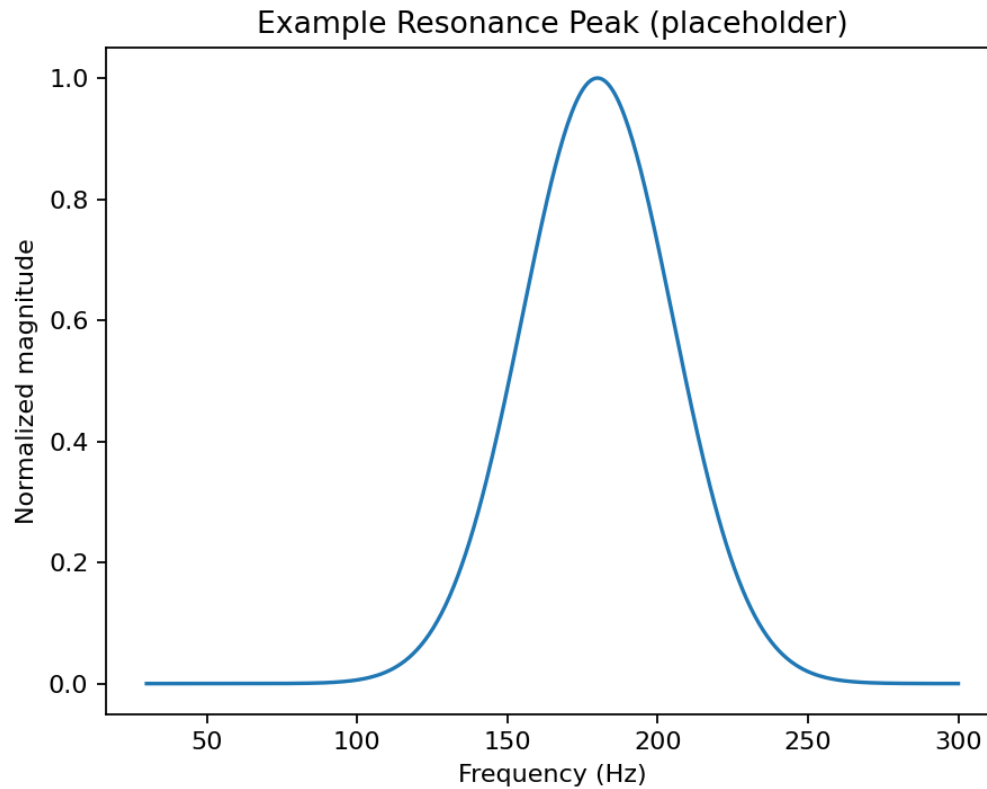
6.2 Resonance Summary (optional)

No resonance_table.json present (this is optional).

6.3 Plots

Selected plots included from bundle/plots:

example_peak.png



7. Error Analysis and Confidence

Uncertainty sources include microphone positioning tolerances, excitation repeatability, environmental reflections (mitigated via close placement and time gating), and support-condition variability. Confidence should be assessed using coherence thresholds and repeated-trial variance (e.g., coefficient of variation across runs).

8. Appendices

8.1 Manifest (if present)

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{ "bundle_version": "0.1", "created_utc": "2025-12-26T00:29:52Z", "hashes": {} }
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